

Red Team Agent – Lab Walkthrough

OWASP Juice Shop | Local LLM (llama3.2) | Kali Linux

Target: `http://localhost:8082`

Overview

This walkthrough documents the exact steps taken to get the `run_agent.py` red team script working end-to-end in the lab environment. Follow each step in order. Screenshots should be inserted where placeholders appear.

Step 1 – Create the Labfiles5.1 Folder and Add Files

Download all the files from the GitHub repository from Labfiles5.1 folder.

Create new folder in the `home/kali` directory and move all downloaded files inside this.

Cut files from downloads folder and paste them inside Labfiles5.1.

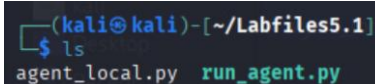
Note that Labfiles5.1 should have only the files and not another folder inside it called Labfiles5.1

Open the terminal and type

```
cd Labfiles5.1
```

Confirm both files are present:

```
ls ~/Labfiles5.1/
```

A terminal window screenshot showing the command `ls` being executed in the directory `~/Labfiles5.1`. The output shows two files: `agent_local.py` and `run_agent.py`.

```
(kali㉿kali)-[~/Labfiles5.1]  
└─$ ls  
agent_local.py  run_agent.py
```

Step 2 – Start OWASP Juice Shop

The target application must be running on port 8082 before the scans can return meaningful results. Pull and start the Juice Shop Docker container.

```
docker run -d -p 8082:3000 --name juice-shop bkimminich/juice-shop
```

```
(kali@kali)-[~/Labfiles5.1]
$ docker run -d -p 8082:3000 --name juice-shop bkimminich/juice-shop
```

Wait approximately 30 seconds for the container to start, then verify it is up:

```
docker ps | grep juice-shop
```

```
(kali@kali)-[~/Labfiles5.1]
$ docker ps | grep juice-shop
5121175dcfcd    bkimminich/juice-shop    "/nodejs/bin/node /j..."    About an hour ago    Up About an hour    0.0.0.0:8082->3000/tcp, [::]:8082->3000/tcp    juice-shop
```

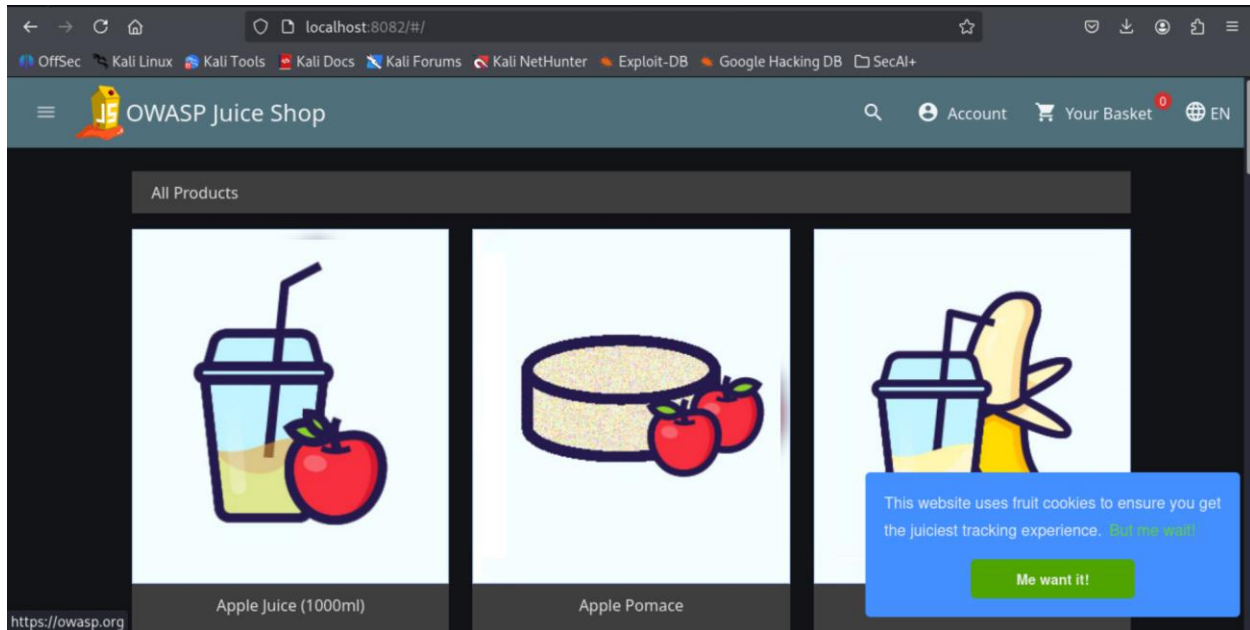
```
curl http://localhost:8082
```

```
(kali@kali)-[~/Labfiles5.1]
$ curl http://localhost:8082
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- SPDX-License-Identifier: MIT

<!doctype html>
<html lang="en" data-beasties-container="OWASP Juice Shop" data-cs="2" data-kind="parent">
  <head>
    <meta charset="utf-8">
    <title>OWASP Juice Shop</title>
    <meta name="description" content="Probably the most modern and sophisticated insecure web application">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <style>@font-face{font-family:'VT323';font-style:normal;font-weight:400;font-display:swap;src:url(https://fonts.gstatic.com/s/vt323/v18/pxiKyp0ihIEF2isQFJXGdg.woff2) format('woff2');unicode-range:U+0102-0103, U+0110-0111, U+0128-0129, U+0168-0169, U+01A0-01A1, U+01AF-01B0, U+0300-0301, U+0303-0304, U+0308-0309, U+0323, U+0329, U+1EA0-1EF9, U+20AB;}@font-face{font-family:'VT323';font-style:normal;font-weight:400;font-display:swap;src:url(https://fonts.gstatic.com/s/vt323/v18/pxiKyp0ihIEF2isRFJXGdg.woff2) format('woff2');unicode-range:U+0100-02BA, U+02BD-02C5, U+02C7-02CC, U+02CE-02D7, U+02DD-02FF, U+0304, U+0308, U+0329, U+1D00-1D0F, U+1E00-1E9F, U+1EF2-1EFF, U+2020, U+20A0-20AB, U+20AD-20C0, U+2113, U+2C60-2C7F, U+A720-A7FF;}@font-face{font-family:'VT323';font-style:normal;font-weight:400;font-display:swap;src:url(https://fonts.gstatic.com/s/vt323/v18/pxiKyp0ihIEF2isFFJU.woff2) format('woff2');unicode-range:U+0000-00FF, U+0131, U+0152-0153, U+02BB-02BC, U+02C6, U+02DA, U+02DC, U+0304, U+0308, U+0329, U+2000-206F, U+20AC, U+2122, U+2191, U+2193, U+2212, U+2215, U+FEFF;}</style>
    <link id="favicon" rel="icon" type="image/x-icon" href="assets/public/favicon_js.ico">
  </head>
  <script>
    window.addEventListener("load", function(){
      window.cookieconsent.initialise({
        "palette": {
          "popup": { "background": "var(--theme-primary)", "text": "var(--theme-text)" },
          "button": { "background": "var(--theme-accent)", "text": "var(--theme-text)" }
        },
        "theme": "classic",
        "position": "bottom-right",
        "content": { "message": "This website uses fruit cookies to ensure you get the juiciest tracking experience.", "dismiss": "Me want it!", "link": "But me wait!", "href": "https://www.youtube.com/watch?v=9PnbKL3wuH4" }
      });
    });
  </script>
</body>
</html>
```

If you get such HTML script response it means it is up

You can also open <http://localhost:8082> in the browser to confirm the Juice Shop UI loads.



Step 3 – Final Successful Run

With Juice Shop running and `agent_local.py` updated, run the script again. All six tools will execute against the live target and the LLM will generate a security report.

```
cd ~/Labfiles5.1
python3 run_agent.py
```

```
(kali@kali)-[~/Labfiles5.1]
$ python3 run_agent.py
```

The script will print progress for each tool as it runs:

- `scan_website` – checks HTTP status and headers
- `discover_api_endpoints` – probes `/api/`, `/rest/`, `/swagger.json`, etc.
- `enumerate_directories` – checks `/admin`, `/login`, `/robots.txt`, `/ftp`, etc.
- `check_admin_endpoints` – checks admin-specific REST paths
- `juiceshop_xss_search_alert` – tests XSS via the search endpoint
- `juiceshop_sql_admin_login_bypass` – tests SQL injection on the login endpoint

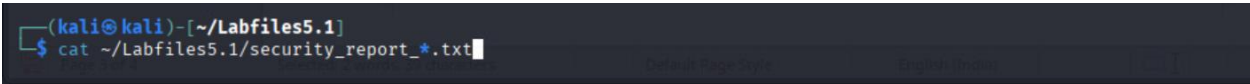
After the tools finish the LLM analyses the results and writes the report. Expected findings:

- SQLi: vulnerable = true (admin login bypass confirmed)
- XSS: payload reflected via search route
- Multiple admin endpoints returning HTTP 200

Step 4 – View the Security Report

The report is saved as security_report_TIMESTAMP.txt in the Labfiles5.1 directory. Open it to review the full findings.

```
cat ~/Labfiles5.1/security_report_*.txt
```

A terminal window with a dark background. The prompt is (kali@kali)-[~/Labfiles5.1]. The command cat ~/Labfiles5.1/security_report_*.txt has been entered and is followed by a cursor.

```
(kali@kali)-[~/Labfiles5.1]  
$ cat ~/Labfiles5.1/security_report_*.txt
```

The report contains: vulnerability summary, accessible paths, raw tool output, AI analysis section, and a high-level AI synopsis (3-5 bullet points suitable for training use).
